

Solvency Field Theory V13.1-V2

How Gravity Works

From irreversible writes to the Schwarzschild metric

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Light bends by the full factor because the same constraint that defines mass — energy at c — also defines the weak-field gravitational metric. The particle and its field are one object. In that sector, $\gamma = 1$ is not a parameter; it is a theorem of one-speed closure.

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1 What this paper claims

SFT derives weak-field general-relativistic gravity — including the full factor-of-two light bending, the Shapiro delay, and Mercury’s perihelion precession — from the mechanics of a locally real particle whose energy propagates at c . In the static spherical sector, the exact Schwarzschild exterior is restored from four capacity axioms, with the horizon interpreted as the surface where outward continuation capacity is exhausted.

SFT does not yet claim: the numerical value of G from first principles; complete gravitational radiation theory; the strong equivalence principle for compact objects; frame-dragging coefficients; Kerr black holes; or dynamic collapse and ringdown. These are registered as open frontier gates with specific pass/fail conditions.

The derivation was rebuilt gate by gate from V12.12 through V13.8 to ensure compatibility with the full locally-real-particle stack.

2 The gates gravity must pass

Any theory that claims to explain gravity faces a gauntlet of precision tests accumulated over a century. Most alternative gravity programs clear the first gate and die at the second. No emergent, entropic, or thermodynamic gravity program has survived them all. Here is where SFT stands on each.

Gate 1: the Newtonian limit. The acceleration around a spherical mass must fall as $1/r^2$, and the acceleration inside a uniform sphere must grow linearly with radius. Nearly every gravity theory passes this gate. SFT passes: the settlement compactness $C_A \propto 1/R$ produces $|\mathbf{a}| \propto 1/R^2$ exterior and $|\mathbf{a}| \propto R$ interior. Retested numerically.

Gate 2: the factor of two. Light passing the Sun is deflected by $1.75''$. A theory with only a scalar potential — one that slows clocks but does not curve space — bends light by half this amount. This is the gate where emergent gravity programs die. Entropic gravity, thermodynamic gravity, and induced-gravity approaches have all stalled here, unable to produce the spatial curvature needed for the full deflection. SFT passes: the one-speed reciprocal closure ($F \times B = 1$) forces spatial stretching to match clock slowing, giving the full factor of two as a theorem, not a fit.

Gate 3: Cassini precision. The Cassini spacecraft measured the space-curvature parameter γ to $|\gamma - 1| < 2.3 \times 10^{-5}$. Any theory whose γ deviates from unity at this level is falsified. SFT passes: $\gamma = 1$ exactly from the one-speed theorem. The Cassini bound is predicted, not merely satisfied.

Gate 4: the Shapiro delay. Radio signals passing near the Sun arrive later than expected by an amount proportional to $1 + \gamma$. With $\gamma = 1$, the delay is twice the Newtonian prediction. SFT passes: the factor $1 + \gamma = 2$ follows from the same reciprocal closure.

Gate 5: perihelion precession. Mercury’s orbit precesses by $43''$ per century beyond the Newtonian prediction. The weak-field precession formula depends on $2 + 2\gamma - \beta$. SFT passes: with $\gamma = \beta = 1$, the precession numerator is exactly 3, matching the GR prediction. Direct timelike orbit integration confirms the result.

Gate 6: exact Schwarzschild. The Schwarzschild metric is the unique static spherical vacuum solution of Einstein’s equations. Many modified gravity theories produce Schwarzschild-like solutions only approximately. SFT passes (bounded): the areal capacity axioms force $A(R) = 1 - 2M/R$ exactly in the static spherical sector.

Gate 7: gravitational wave structure. LIGO observes two tensor polarizations propagating at the speed of light. A scalar theory generically produces a third “breathing” polarization and may predict the wrong propagation speed. SFT status: the trace-free settlement-lane channel produces

the correct two-polarization skeleton, but scalar breathing-mode suppression is not yet proven. *Frontier*.

Gate 8: the binary pulsar. The Hulse–Taylor binary pulsar loses orbital energy at a rate matching GR’s quadrupole prediction to 0.1%. Scalar theories generically produce forbidden dipole radiation that would drain energy faster than observed. SFT status: universal write charge provides a natural suppression route for dipole radiation, but the proof is not yet complete for compact self-gravitating objects. *Frontier*.

Gate 9: frame dragging. Gravity Probe B and LAGEOS measured the dragging of inertial frames by Earth’s rotation. A scalar potential has no mechanism for this effect. SFT status: the mixed continuation–settlement channel carries the correct topology for rotating-source effects, but the Lense–Thirring coefficient is not yet derived. *Frontier*.

SFT clears gates 1 through 6. Gates 7 through 9 have native structural homes — the lane channels — but their detailed physics remains open. The honest summary: SFT’s weak-field and static-spherical gravity is earned. The radiation, equivalence, and frame-dragging sectors are frontier work with identified paths but no sealed results.

3 The particle that makes the field

In SFT, a massive particle is energy caught in a closed orbit at speed c . The energy cannot stop. It cannot exceed c . It cannot overwrite what it has already written. Each tick of the particle’s Compton clock writes one irreversible settlement increment to the local geometry. The write rate is set by the energy:

$$\dot{N} = \frac{E}{\hbar} \quad (\text{reduced/radian gauge}). \quad (1)$$

Every write deposits one universal footprint of area a_ϕ . This footprint does not depend on the particle species, the local geometry, or the amount of settlement already present. One write, one footprint, always.

The accumulated settlement at distance R from a static source of energy E is the compactness:

$$C_A(E, R) = \frac{a_\phi E}{\hbar c R}. \quad (2)$$

With the identification $G_{\text{eff}} = a_\phi c^3/\hbar$, this becomes $C_A = G_{\text{eff}} M/(c^2 R)$ for rest energy $E = Mc^2$, and the acceleration readout $\mathbf{a} = -c^2 \nabla C_A$ recovers the Newtonian inverse-square law: $|\mathbf{a}| = G_{\text{eff}} M/R^2$.

Radiation gravitates with double weight relative to cold matter because a photon commits its entire c -budget to outward spatial propagation, while a massive particle divides the same budget between internal circulation and external motion. The active source density for a relativistic fluid is $\rho_{\text{active}} = (\varepsilon + 3p)/c^2$, giving a source factor of $1 + 3w$: one for cold matter ($w = 0$) and two for radiation ($w = 1/3$).

4 Why the scalar objection fails

The compactness C_A is a scalar. A scalar potential slows clocks but does not curve space. Light moving through a scalar field is deflected only by the time-curvature component, giving half the observed bending. This is the graveyard of emergent gravity: entropic, thermodynamic, and induced-gravity programs have all stalled or died at this point.

SFT survives because settlement is not only a scalar.

The projector partition of V13.8 says that 3+1 write capacity splits into three settlement lanes and one continuation lane. Settlement deposits into all three spatial lanes. The scalar C_A is the *trace* of this lane structure — the Newtonian/00 projection. The trace-free part carries quadrupole shear (the skeleton of gravitational radiation), and the mixed continuation–settlement sector carries circulation/orientation information (the skeleton of frame dragging). SFT’s gravitational object is not a scalar potential. It is a lane-structured settlement field whose scalar projection gives Newtonian gravity and whose tensor structure gives the rest.

5 The one-speed theorem and $\gamma = 1$

The founding axiom of SFT is that real support propagates at one invariant speed c . This axiom defines the particle: energy at c caught in closure. The same axiom defines the gravitational metric.

Settlement slows the local clock. Let the clock-rate factor be $F = e^{-q_t \phi}$, where ϕ is the dimensionless settlement depth. A particle moving through this settlement must still move at total speed c locally. If the clock runs slow, the spatial scale *must* stretch to compensate, so that the locally measured speed of light remains c . The spatial scale factor is $B = e^{+q_s \phi}$.

The one-speed budget requires:

$$F B = 1 \implies e^{(q_s - q_t)\phi} = 1 \text{ for all } \phi \implies q_s = q_t. \quad (3)$$

Define $\alpha = q_s/q_t$. The one-speed budget forces $\alpha = 1$. This is not a selection from a menu of alternatives. It is a theorem: any other value of α would change the local speed of light in a settlement field, violating the founding axiom.

With $\alpha = 1$:

$$F = e^{-\phi}, \quad B = e^{+\phi}, \quad F B = 1. \quad (4)$$

The weak-field expansion gives the post-Newtonian parameters:

$$g_{00} = -F^2 \approx -(1 - 2\phi), \quad (5)$$

$$g_{ij} = +B^2 \delta_{ij} \approx (1 + 2\phi) \delta_{ij}. \quad (6)$$

Reading off the standard PPN coefficients:

$$\boxed{\gamma = 1, \quad \beta = 1, \quad 1 + \gamma = 2, \quad 2 + 2\gamma - \beta = 3.} \quad (7)$$

Light and radio signals are deflected by both the time-curvature term and the space-curvature term. With $\gamma = 1$, both terms contribute equally, giving the full factor-of-two deflection measured since the 1919 eclipse and confirmed by the Cassini spacecraft to $|\gamma - 1| < 2.3 \times 10^{-5}$.

Direct numerical integration of null (photon) and timelike (planetary) trajectories through the exponential closure confirms the weak-field results: the null deflection ratio to GR is 1.003, and the timelike perihelion precession ratio is 0.998, both within integration tolerance.

6 What sets the strength of gravity

The gravitational coupling takes the form:

$$G_{\text{eff}} = \frac{a_\phi c^3}{\hbar}, \quad A_0 = 2\pi a_\phi, \quad (8)$$

where a_ϕ is the per-radian root write area and A_0 is the full-cycle write area. The 2π connecting them is the reduced-to-full-cycle gauge conversion earned from Compton phase closure.

All faces of the root invariant — root area, root length, root time, root density, root tension — form one equivalence class. Supplying any single face determines all others and fixes G .

SFT does not derive the numerical value of a_ϕ from first principles. The root-area invariant compresses the normalization problem to one number, but that number remains an empirical input. The Planck area $\ell_P^2 = G\hbar/c^3$ cannot be used as a derivation of G because it already contains G ; using it would be circular. This is stated as an explicit open gate.

7 The galaxy bridge

At scales much larger than the source, the settlement acceleration scale

$$a_S = \frac{cH_0}{2\pi} \approx 1.05 \times 10^{-10} \text{ m/s}^2 \quad (9)$$

sets the boundary between the Newtonian regime and the settlement-dominated regime. This scale emerges from the same Compton-phase-closure mechanics that give the particle its mass; it is not fitted to galaxy data.

The SPARC galaxy rotation-curve program (V13 Fork 08) uses a participation-weighted settlement model with zero adjustable parameters and outperforms the best-fitted global constant on 175 galaxies at 3389 radial points. The local-metric closure ($\gamma = 1$, $\beta = 1$) is explicitly protected: the far-field participation operator λ_{eff} and the scale a_S live in the galaxy-support sector, not the local metric sector.

At high redshift, the scale $a_S \propto H(z)$ rises. The theory registers a falsifiable inequality: high- z galaxies can show weaker settlement boosts than local galaxies only if the participation factor $\lambda_{\text{eff}}(z)$ falls faster than $H(z)$ rises.

8 Exact Schwarzschild from capacity axioms

In the static spherical sector, the metric can be written in areal coordinates:

$$ds^2 = -A(R) dt^2 + A(R)^{-1} dR^2 + R^2 d\Omega^2, \quad (10)$$

where the areal radius R satisfies $\text{Area} = 4\pi R^2$. The lapse function $A(R)$ is identified with the remaining continuation capacity at shell radius R : the fraction of outward write addresses not yet consumed by accumulated settlement.

Four axioms fix $A(R)$:

Areal-shell nativity. A spherical write boundary is measured by physical area, making R the native shell variable.

Literal continuation capacity. $A(R)$ is the remaining fraction of outward write addresses at radius R .

Universal footprint. Each write costs one footprint regardless of existing settlement depth. The marginal capacity cost is constant: $dK/dC = \text{const}$.

Additive capacity. Independent source packages do not create settlement–settlement cross terms: $K(C_1 + C_2) = K(C_1) + K(C_2)$.

The general one-parameter family of capacity functions consistent with weak-field behavior is:

$$A_q(C) = 1 - 2C + qC^2, \quad C = M/R. \quad (11)$$

Universal footprint requires $dK/dC = 2 - 2qC$ to be constant, forcing $q = 0$. Independently, additivity requires $K(C_1+C_2) - K(C_1) - K(C_2) = -2q C_1 C_2 = 0$ for all C_1, C_2 , again forcing $q = 0$.

Two routes. Same conclusion:

$$\boxed{A(R) = 1 - \frac{2M}{R}}. \quad (12)$$

This is the exact Schwarzschild exterior. The horizon at $R = 2M$ is where $A = 0$: continuation capacity is exhausted. Every outward-pointing write address at that shell has been consumed by settlement. No outward continuation remains. This is the SFT interpretation of the event horizon — not a coordinate artifact, but a write-capacity boundary.

Boundary: this result is bounded to the static spherical areal-capacity sector. Kerr (rotating black holes), dynamic collapse, ringdown, horizon microphysics, and black-hole thermodynamics remain open.

9 Frontiers

Three frontier gates remain armed. Each has a native SFT home and a specific kill condition.

Gravitational radiation

The trace-free settlement-lane channel admits a transverse-traceless far-zone readout with two polarization degrees of freedom (plus and cross) and quadrupole $m = 2$ source structure. The tensor-wave skeleton is viable, but the dangerous channels are not yet closed: scalar breathing-mode suppression, dipole radiation suppression, the quadrupole power coefficient, and the binary-pulsar energy-loss rate. The kill condition is Hulse–Taylor-level orbital decay.

Equivalence and dipole suppression

SFT’s native route to the equivalence principle is universal write charge: gravity couples to write-rate energy, not to material composition. If the gravitational charge per inertial mass $\kappa_i = q_i/m_i$ is the same for all bodies — including their gravitational binding energy — then free-fall universality holds and binary dipole radiation is suppressed. The open question is whether compact-object binding energy and gravitational self-energy maintain universal κ . A clue from the particle program suggests they should: neutron stars may represent a transition from ordinary gravitational binding toward the same self-maintaining closure mechanism that defines particles, implying the same coupling by construction.

Frame dragging

The mixed continuation–settlement channel carries rotating-source effects with the correct topology: the circulation field falls as r^{-2} , the frame-drag precession as r^{-3} , the axis-to-equator ratio is 2:1, and spin reversal flips the sign. The Lense–Thirring coefficient, gyroscope transport law, and Kerr metric are not yet derived.

10 The sentence

Gravity is not a force added to the particle. Gravity is the accumulated settlement of irreversible writes in 3+1 spacetime. Settlement slows clocks. The one-speed budget stretches rulers by the reciprocal amount. The same axiom that defines mass defines the

metric. Light bends by the full factor because the particle and its field are one object, and that object moves at c .

11 Claim boundary

Claim	Status	Boundary
Newtonian scalar projection	Earned	$C_A \propto 1/R$, $ \mathbf{a} \propto 1/R^2$. Full gravity is not scalar alone.
$\gamma = 1$, full light bending	Earned (weak field)	Theorem of one-speed closure. Does not cover strong-field lensing.
Structure of G	Earned	$G = a_\phi c^3/\hbar$; numerical a_ϕ open.
Schwarzschild exterior	Earned (static spherical)	From capacity axioms. No Kerr, ringdown, or dynamics.
Horizon	Earned (interpretation)	Continuation-capacity exhaustion at $R = 2M$. No microphysics or thermodynamics.
Gravitational waves	Frontier	TT skeleton viable. Scalar/dipole/pulsar open.
Equivalence principle	Frontier	Universal κ route. Binding energy and compact objects open.
Frame dragging	Frontier	Topology earned. Coefficient and transport open.

Acknowledgments

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References

- [1] S. Fosmark, *Solvency Field Theory V13*, Zenodo (2026). <https://zenodo.org/records/20822219>